MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, HIGH TEMPERATURE AND FIRE RESISTANT, LIGHT WEIGHT, SMALL DIAMETER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the wire described herein shall consist of this specification and the latest issue of MIL-W-25038.

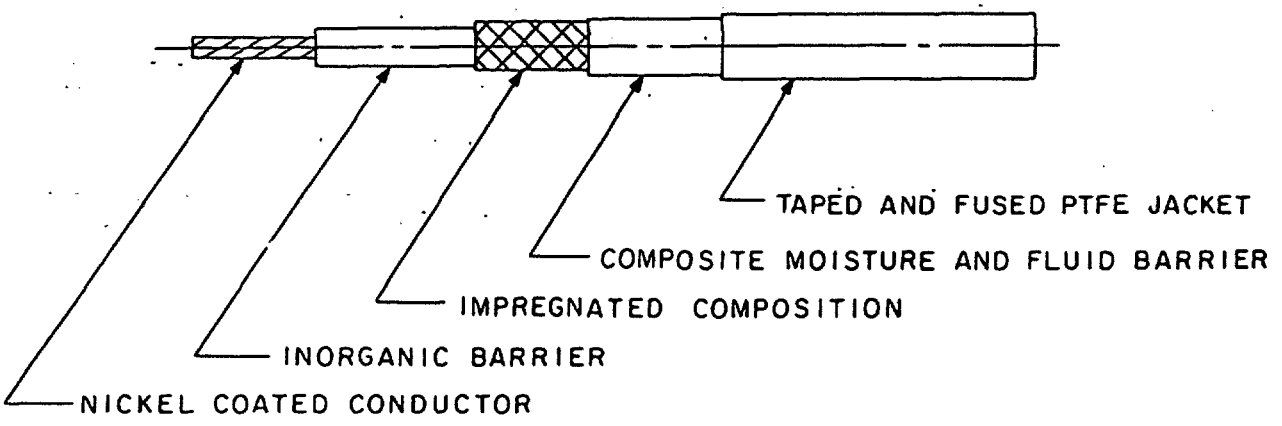


FIGURE 1. Wire, electrical.

REQUIREMENTS:

Design and construction: See figure 1.

Temperature rating: 260°C.

Voltage rating: 600 V rms.

Thermal shock: 0.60 inch maximum change in insulation measurement.

Color: Unless otherwise specified by the part number, the color shall be white.

Part number: M25038/3- (wire size from table I) (color code designator in accordance with MIL-STD-681).

Examples:
Size 20 AWG, white = M25039/3-20-9.
Size 20 AWG, white with orange stripe = M25038/3-20-93.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:26Z
Check the source to verify that this is the current version before use.

TABLE I. Finished wire construction and performance details.

Contact wire barrel size 4/ 5/	Wire size 1/ 2/	Conductor strand 3/			Maximum condition diameter (inch)	Maximum dc resistance 1,000 ft. at 20°C (ohms)	Finished wire diameter (max) (inch)	Weight 1,000 ft. (pounds) (max)	Performance details	
		No.	AWG	Nominal diameter					Bend test mandrel OD (inches)	Bend test mandrel Load (pounds)
16	20	19	32	.008	.041	15.27	.093	9.0	6	2
16	18	19	30	.010	.052	8.5	.103	11.5	6	2
16	16	19	29	.0113	.061	6.66	.117	14.5	8	3

1/ Wire size 20 shall consist of nickel-coated, high-strength copper conductor.
2/ Wire sizes 18 and 16 shall consist of nickel-coated, copper conductor.
3/ The conductor strands shall have an annular nickel coating, which its weight shall not be less than 27 percent of the total weight of the nickel-coated strand when tested in accordance with ASTM B355.
4/ Size 16 contact may be used with all three wire sizes which will increase the wire sealing range of the connector.
5/ Special wire stripping procedures: Strip the oversized wire .1875 inch to .25 inch beyond the end of the crimp barrel. This extra length is still below the wire seals and will allow use of the standard insertion and removal tool.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-0900)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:26Z
Check the source to verify that this is the current version before use.